MICROSOFT AZURE SQL

MODULE: APPLICATION DEVELOPMENT 3B

CONTENT DEVELOPED BY: CASSIM VANKER

LEARNING OBJECTIVES

- Microsoft Azure SQL
  - What is a Database as Service?
  - Difference between SQL Server and Azure SQL
  - What is Azure SQL?
  - How do I create an Azure SQL DB?
  - Create a server-level firewall rule
  - Georeplication
  - Georeplication: Failover
  - Transparent Data Encryption
  - Alert Rules
  - Performance Overview
  - Automatic Tuning

WHAT IS DATABASE AS A SERVICE?

- A database service is a public cloud offering that has 3 characteristics:
  - The service provider owns and manages the database software, common administration tasks like backups, high availability, the operating system and bare metal hardware.
  - The user pays according to the usage of the service
  - Users can scale up or down on demand and create or destroy environments with no service provider intervention.

DIFFERENCE BETWEEN SQL SERVER AND AZURE SQL

Name	Microsoft Azure SQL Database	Microsoft SQL Server
Description	Database as a Service offering with high compatibility to Microsoft SQL Server	Microsoft relational DBMS
Primary database model	Relational DBMS	Relational DBMS
Secondary database models	Document store	Document store
	Graph DBMS	Graph DBMS
	Key-value store	Key-value store
Cloud-based info	yes	no
Implementation language	C++	C++
Server operating systems	hosted	Linux
		Windows
Data scheme	yes	yes
Typing info	yes	yes
XML support info	yes	yes
Secondary indexes	yes	yes
SQL info	yes	yes
APIs and other access methods	JDBC	OLE DB
	ODBC	Tabular Data Stream (TDS)
	ADO.NET	ADO.NET
		JDBC
		ODBC

DIFFERENCE BETWEEN SQL SERVER AND AZURE SQL

Name	Microsoft Azure SQL Database	Microsoft SQL Server
Supported programming languages	.Net	C#
	C#	C++
	Java	Delphi
	JavaScript (Node.js)	Go
	PHP	Java
	Python	JavaScript (Node.js)
	Ruby	PHP
		Python
		R
		Ruby
		Visual Basic
Server-side scripts info	Transact SQL	Transact SQL and .NET languages
Triggers	yes	yes
Partitioning methods info		tables can be distributed across several files (horizontal partitioning); sharding through federation
Replication methods info	yes, with always 3 replicas available	yes, but depending on the SQL-Server Edition
Distributed Transactions	No	Yes

OTHER DIFFERENCE BETWEEN SQL SERVER AND AZURE SQL

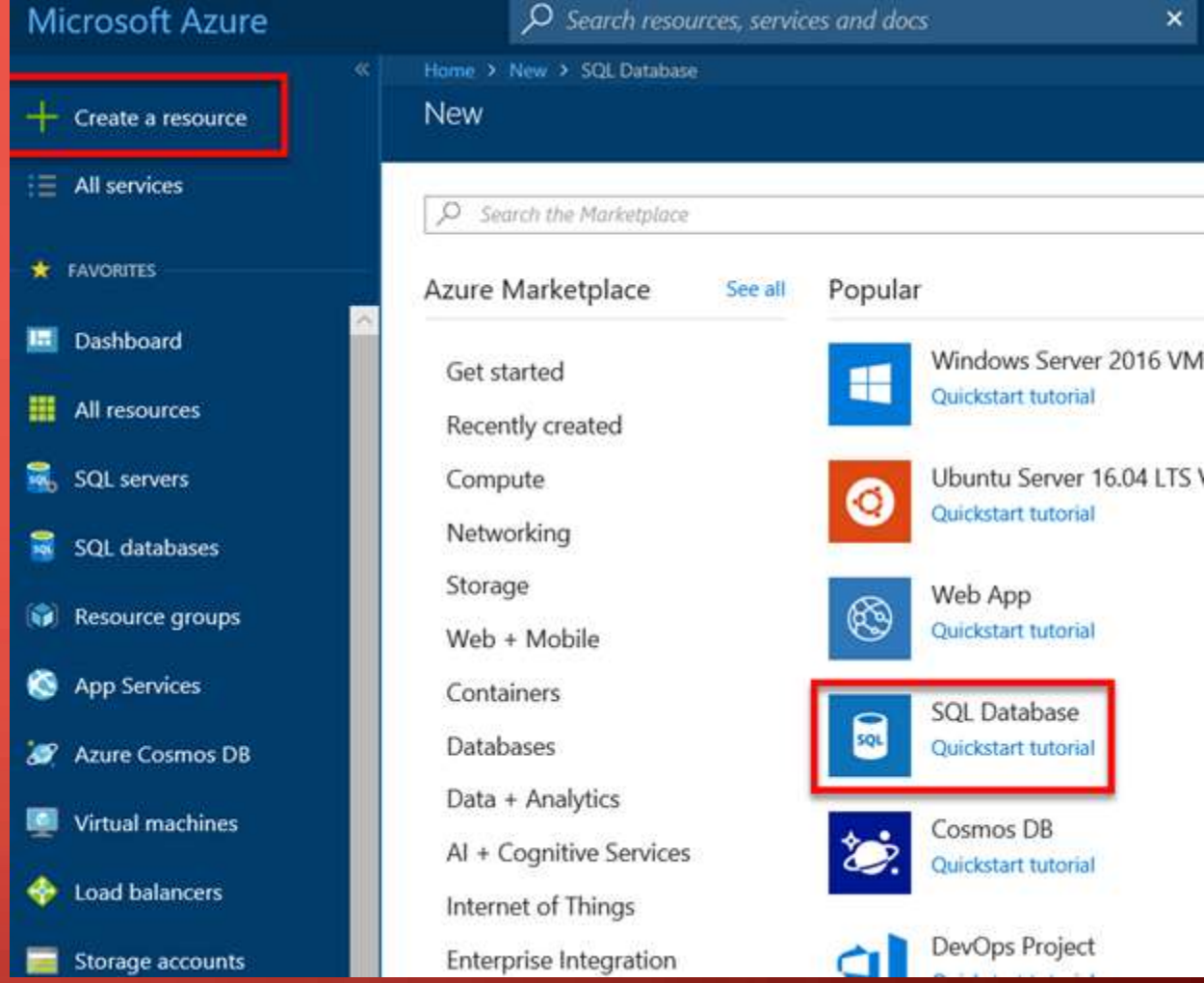
- Few features have full parity with the SQL Server;
- Distributed transactions are not supported.
- Agented jobs using SQL Agent are not supported.
- Extremely large connections (>2GB logging) could be throttled or disconnected

WHAT IS AZURE SQL?

- A database as a service platform.
- Offered inside Microsoft Azure.
- Key capabilities include:
  - Continuous learning of your unique app patterns, adaptive performance tuning, and automatic improvements to reliability and data protection.
  - Scaling as needed, with virtually no app downtime.
  - Management and monitoring of multitenant apps with isolation benefits of one-customer-per-database
  - Leverage open source tools
  - Data protection with encryption, authentication, limiting user access to the appropriate subset of the data, continuous monitoring and auditing to help detect potential threats and provide a record of critical events in case of a breach
- User can use the same tools and frameworks to query and insert Azure SQL

HOW DO I CREATE AN AZURE SQL DB?

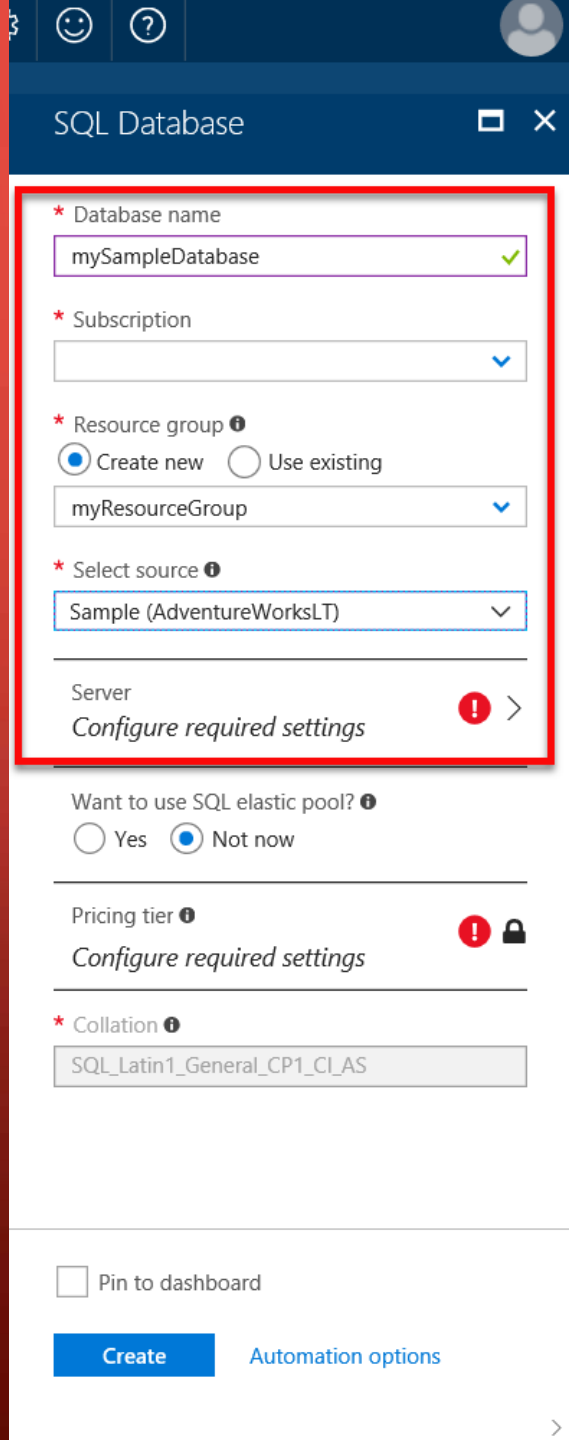
- Login to the Azure Portal
- Click Create a resource in the upper left-hand corner of the Azure portal.



HOW DO I CREATE AN AZURE SQL DB?

- Fill out the SQL Database form with the following information, as shown on the preceding image:

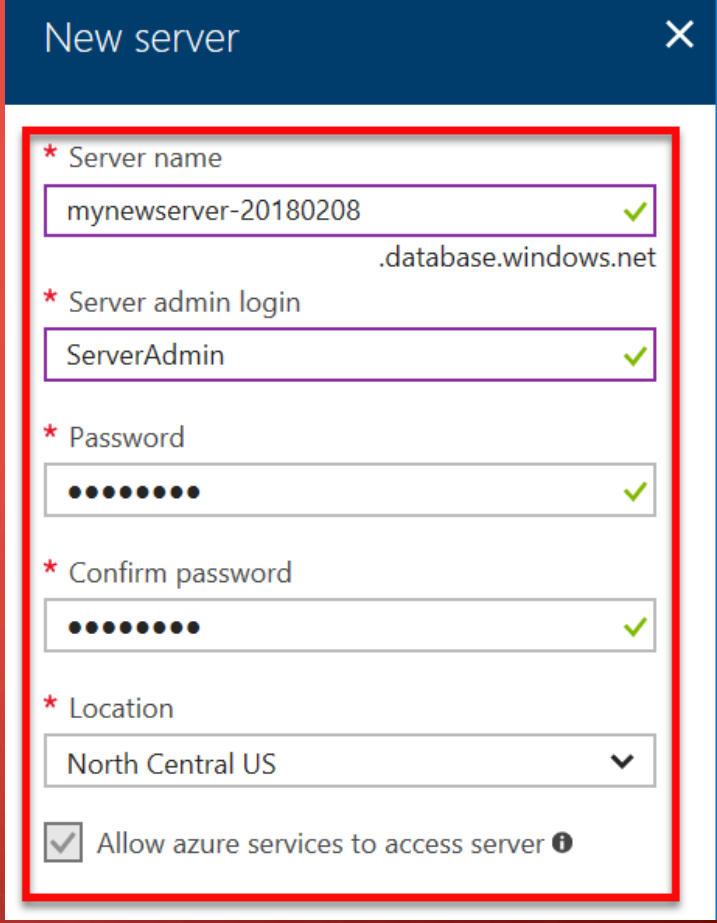
Setting	Suggested value
Database name	myDUTDatabase
Subscription	My Subscription
Resource group	Group name for this resource
Select source	Select a Data Source: Blank Database Sample Backup



HOW DO I CREATE AN AZURE SQL DB?

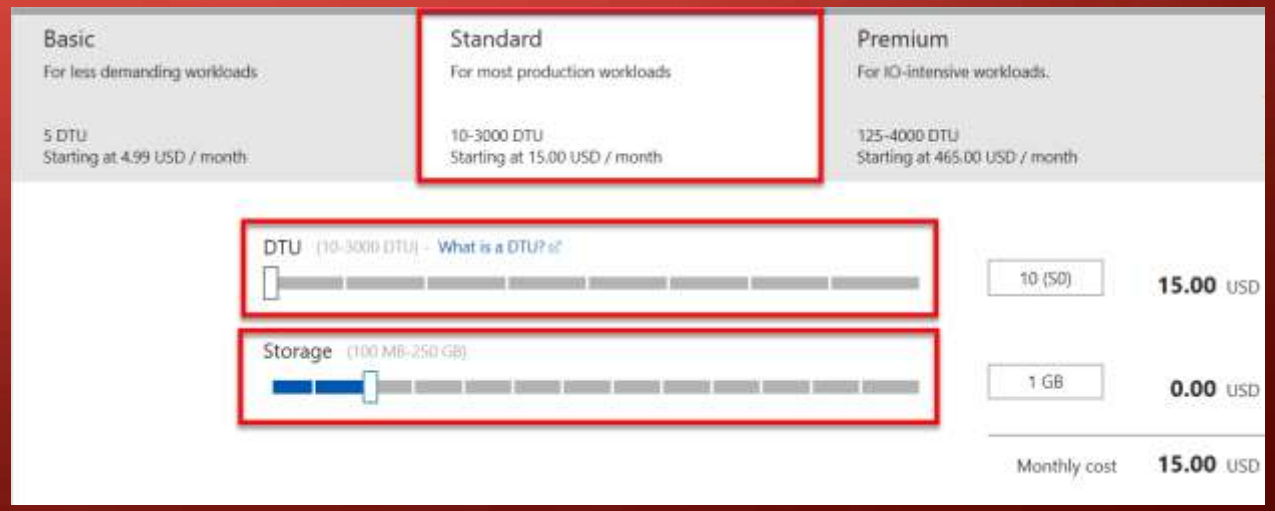
- Under **Server**, click **Configure required settings** and fill out the SQL server (logical server) form with the following information, as shown on the following image.
- A server is a container of Azure SQL DB's

Setting	Suggested value
Server name	Any globally unique name
Server admin login	Any valid name
Password	Any valid password
Subscription	Your subscription
Resource group	myDUTGroup
Location	Any valid location



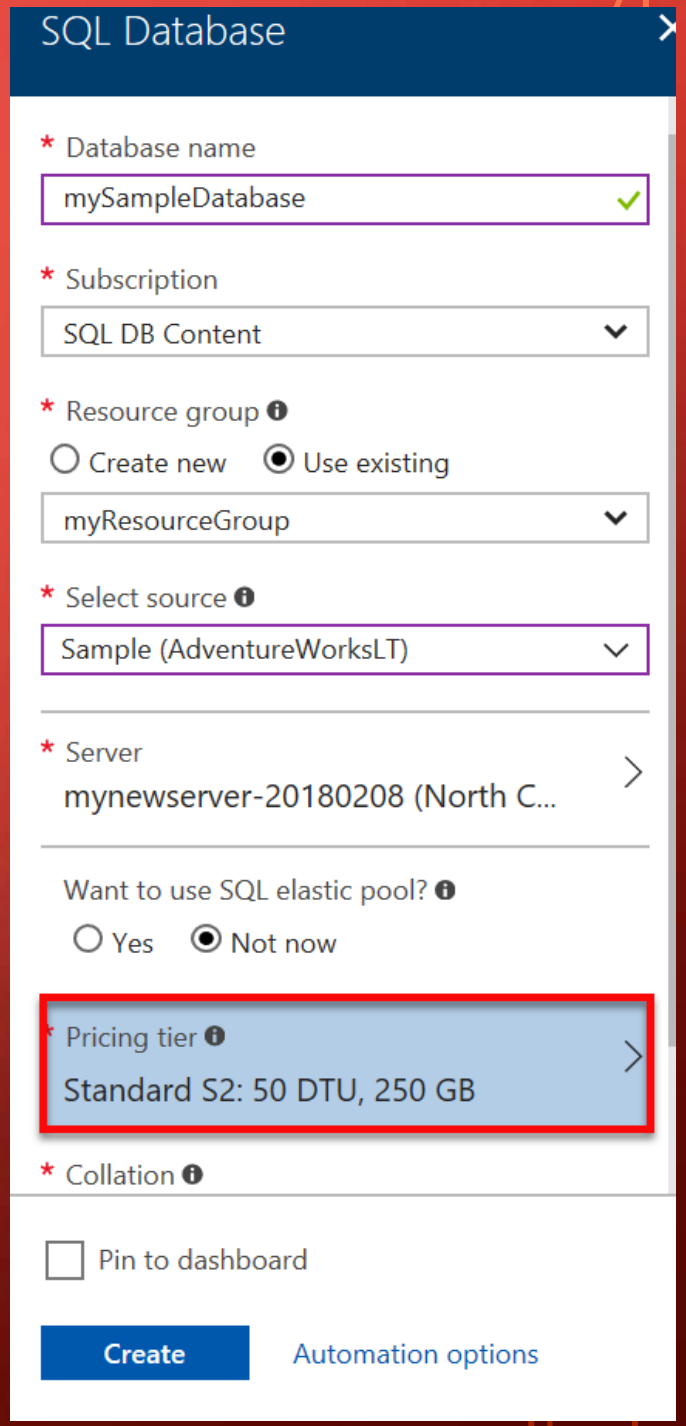
HOW DO I CREATE AN AZURE SQL DB?

- Click Pricing tier to specify the service tier, the number of DTUs (database transactional unit – mix of cpu, memory and IO), and the amount of storage. Explore the options for the amount of DTUs and storage that is available to you for each service tier.
- Select the appropriate service tier and then use the slider to select DTUs and storage, click **Apply**.



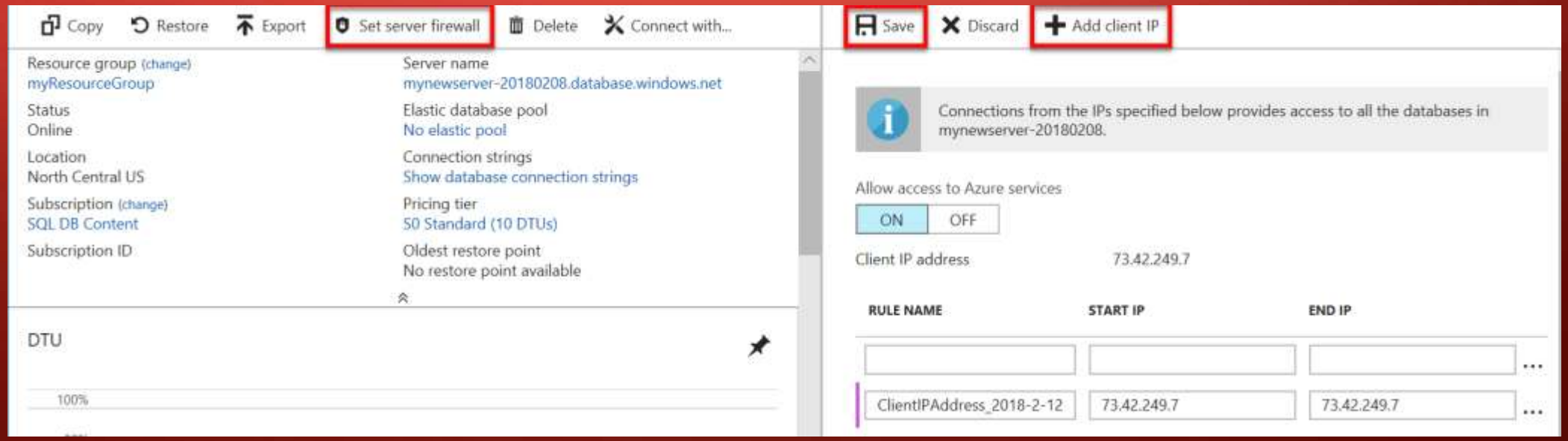
HOW DO I CREATE A AZURE SQL DB?

- Now that you have completed the SQL Database form, click **Create** to provision the database. Provisioning takes a few minutes.



CREATE A SERVER-LEVEL FIREWALL RULE

- After the deployment completes, click SQL databases from the left-hand menu and then click on the database on the SQL databases page.
- The overview page for your database opens, showing you the fully qualified server name. Click set server firewall, then click Add Client IP and Save

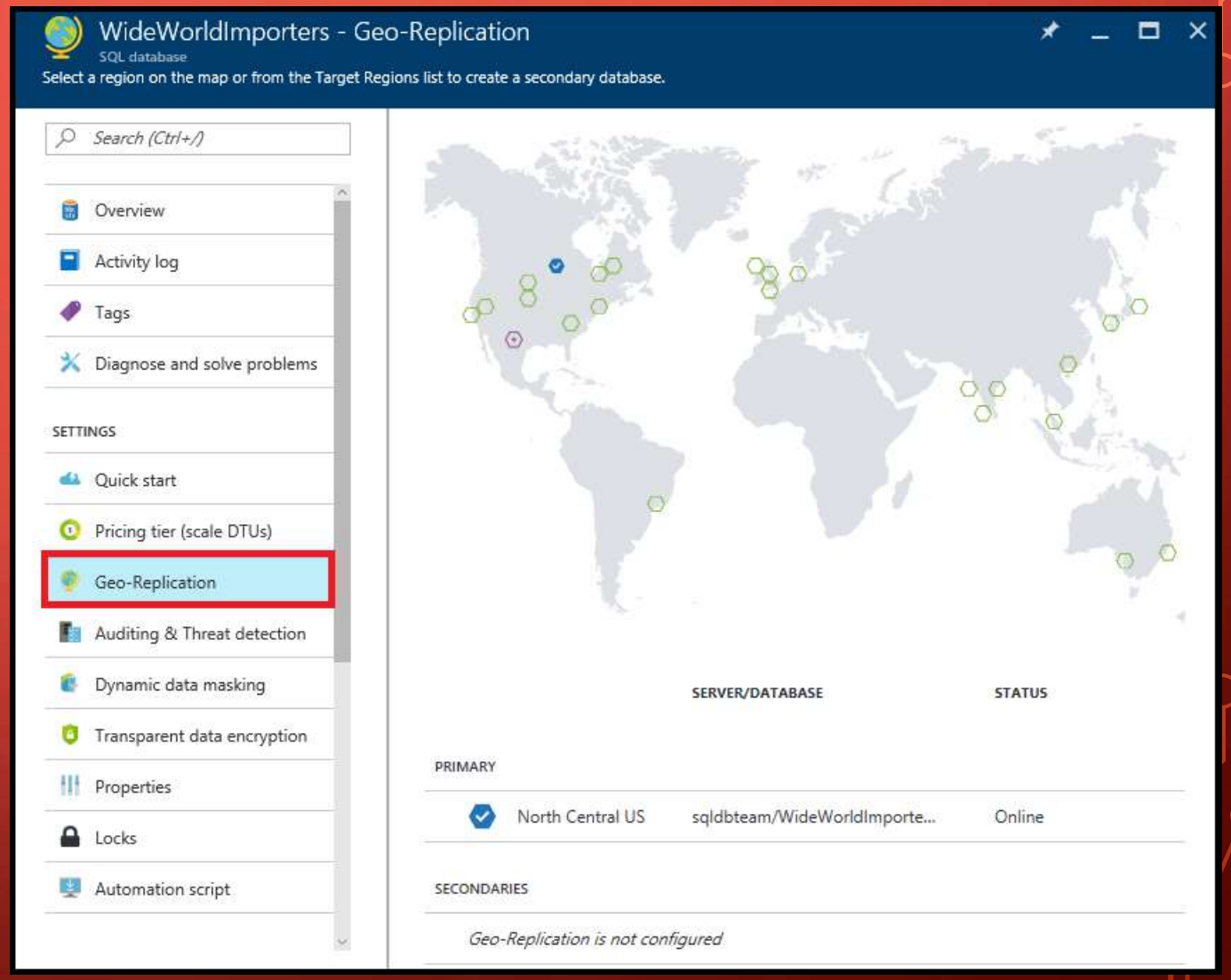


GEOREPLICATION

- To add a secondary database, you must be the subscription owner or co-owner.
- On the SQL database page, select geo-replication, and then select the region to create the secondary database. You can select any region other than the region hosting the primary database. A paired region is recommended.
- A list of paired regions can be found here:

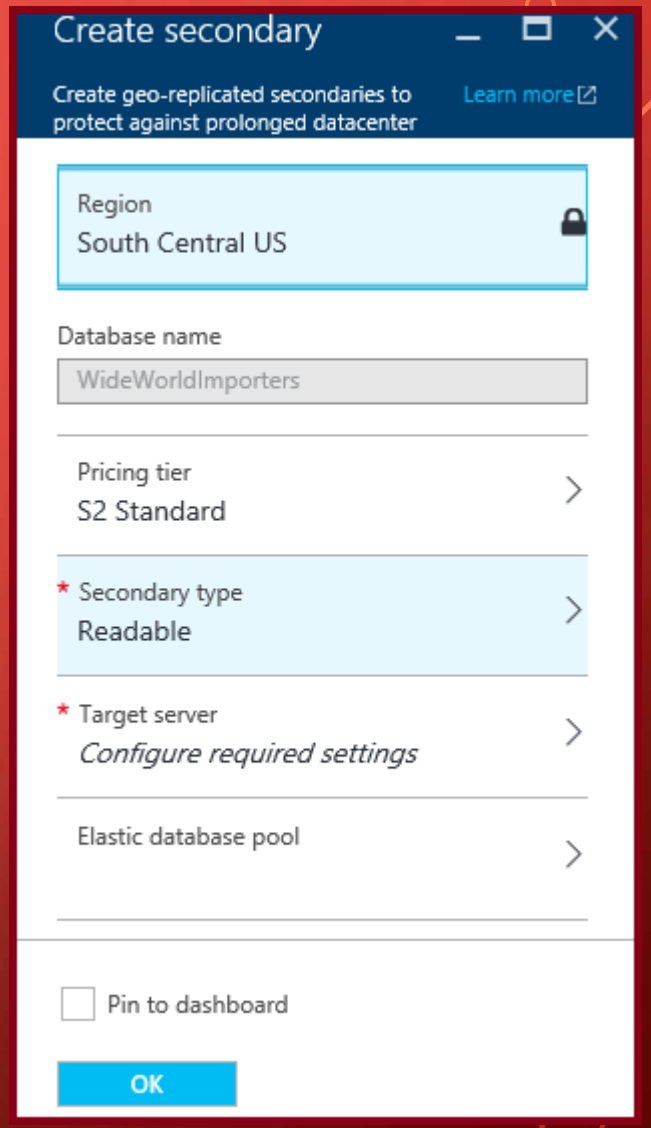
https://docs.microsoft.com/en-us/azure/best-practices-availability-paired-regions

GEOREPLICATION



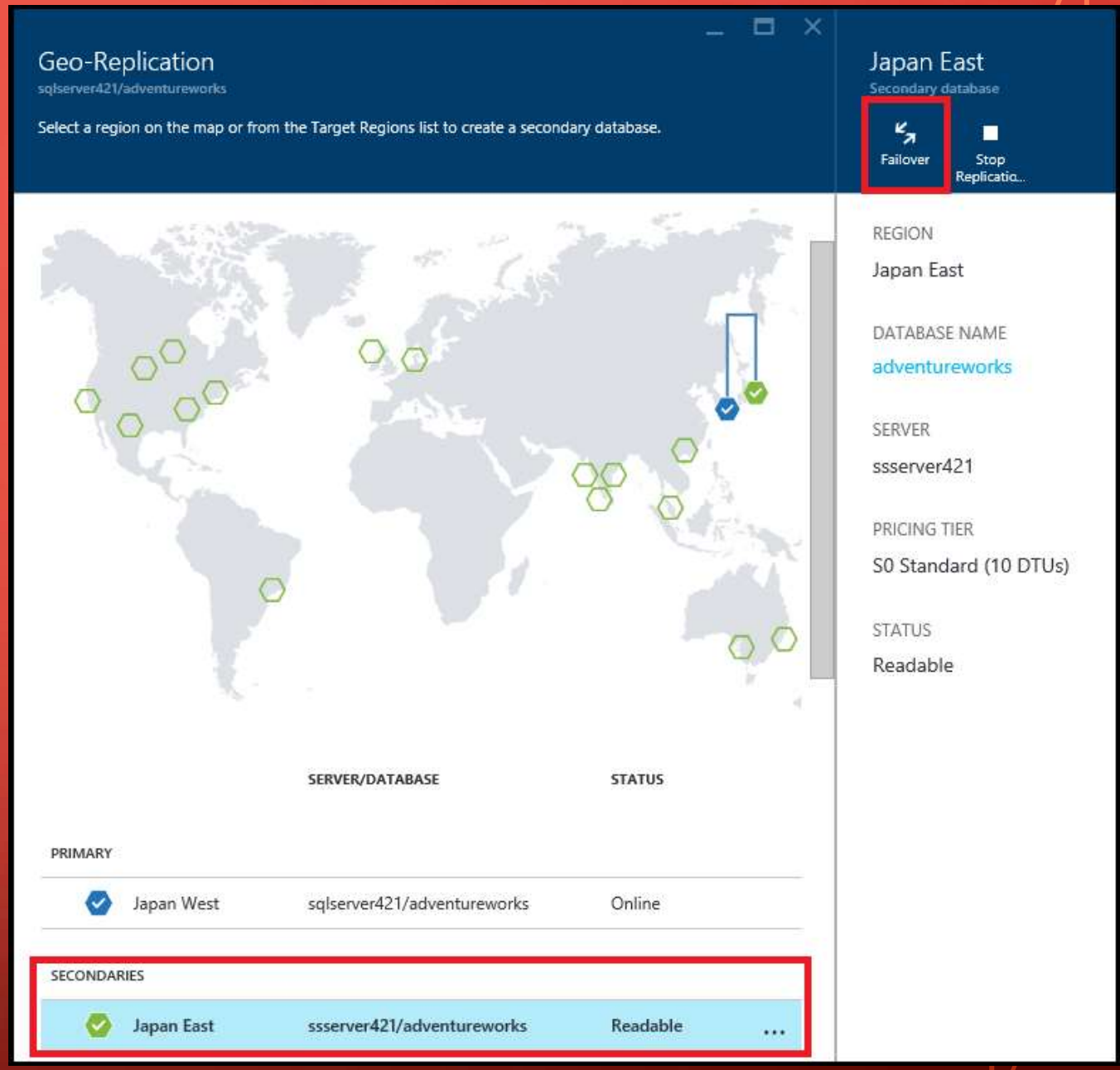
GEOREPLICATION

- Select or configure the server and pricing tier for the secondary database.
- Click **Create** to add the secondary.
- The secondary database is created and the seeding process begins.



GEO-REPLICATION: FAILOVER

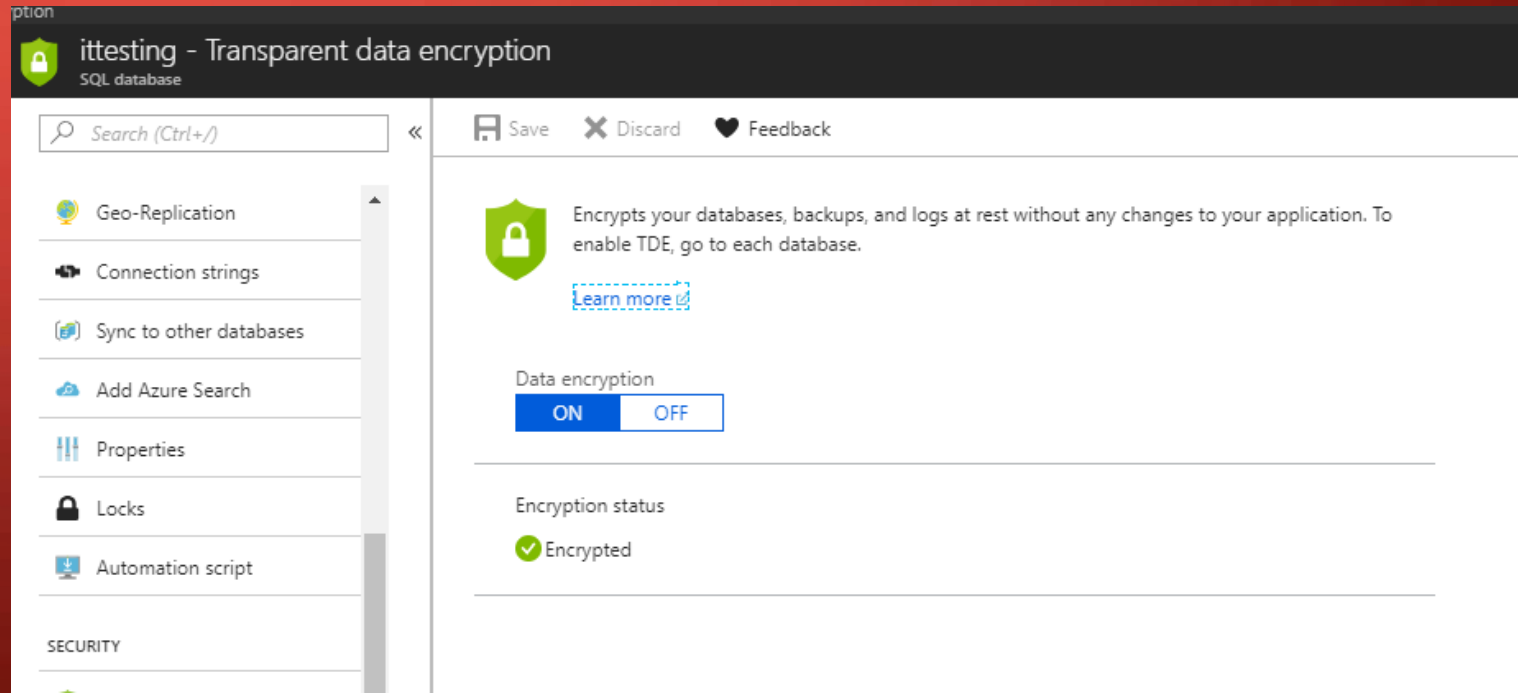
- On the SQL Database blade select **All settings** > **geo-replication.**
- In the **SECONDARIES** list, select the want to become the new primary and click **Failover.**



TRANSPARENT DATA ENCRYPTION

- If you are developing an application that requires an encrypted database connection.
- You can turn on transparent data encryption for your Azure SQL database.
- To enable:

Turn on and select save.

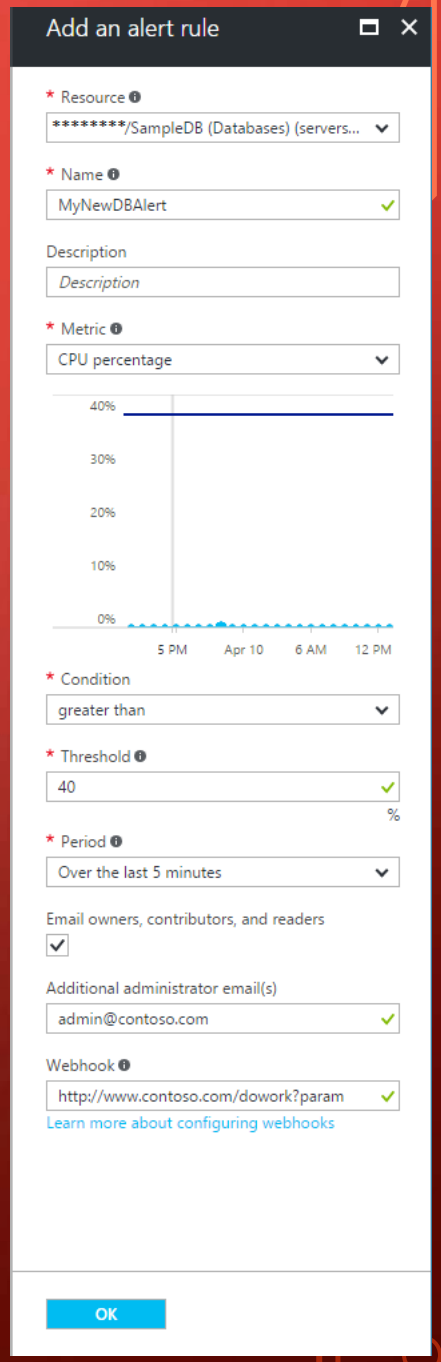


ALERT RULES

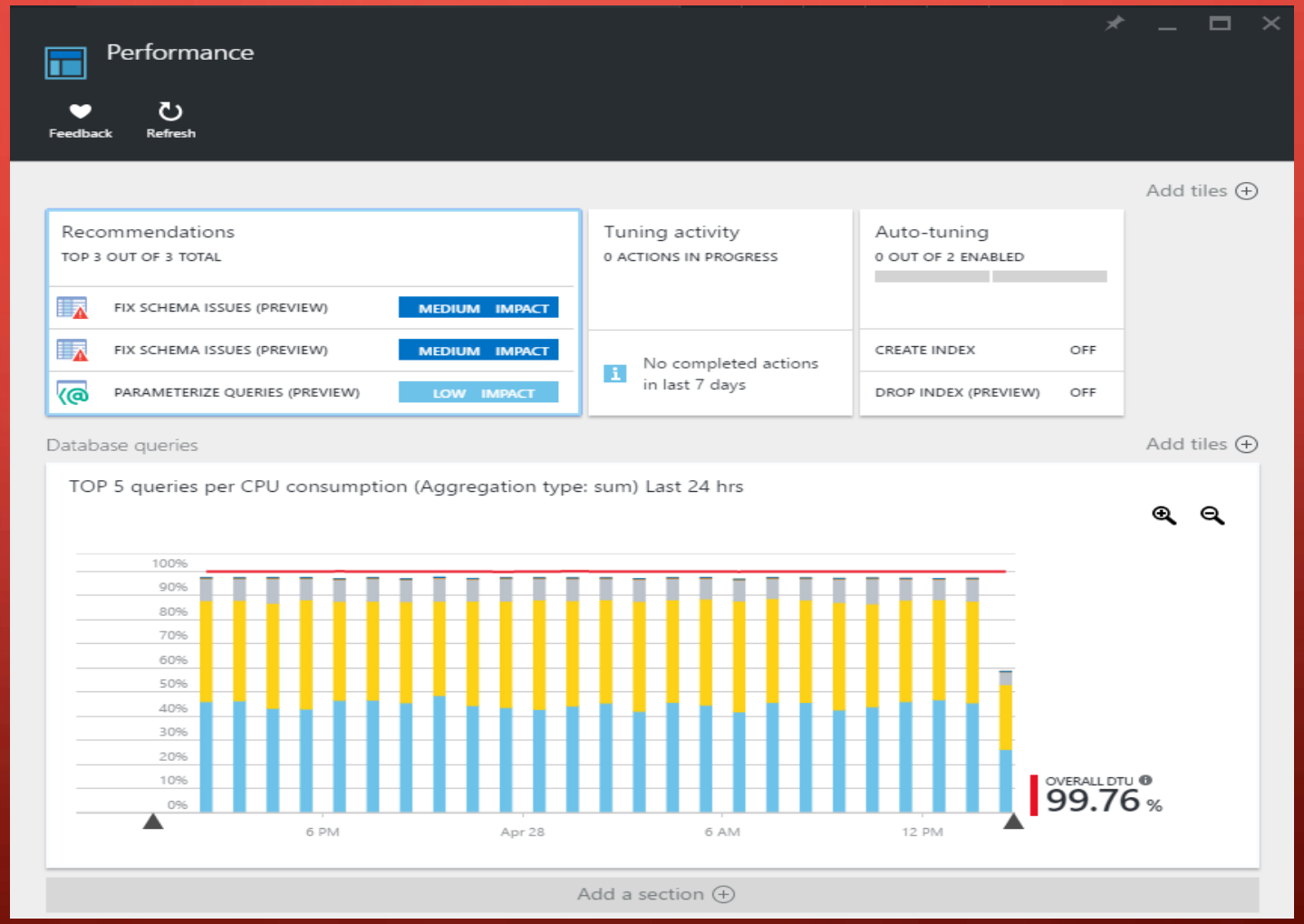
- You can receive an alert based on monitoring metrics for, or events on, your Azure services.
  - **Metric values** - The alert triggers when the value of a specified metric crosses a threshold you assign in either direction. The event triggers both when the condition is first met and then afterwards when that condition is no longer being met.
  - **Activity log events** - An alert can trigger on *every* event, or, only when a certain number of events occur.
- You can configure an alert to do the following when it triggers:
  - send email notifications to the service administrator and co-administrators
  - send email to additional emails that you specify.
  - call a webhook

ALERT RULES

- In the portal, choose the SQL Server you want to add rules for
- On the SQL database page, Select **Alerts** or **Alert rules** under the MONITORING section.
- Click Add Alert (windows on right shows)
- Select the matric that you want to monitor, the condition, threshold and Period.
- Select the action (email/Webhook)
- Save the Alert.



PERFORMANCE MONITORING



PERFORMANCE MONITORING

- This view provides a summary of your database performance, and helps you with performance tuning and troubleshooting.
- The **Recommendations** tile provides a breakdown of tuning recommendations for your database (top three recommendations are shown if there are more). Clicking this tile takes you to **Performance recommendations.**
- The **Tuning activity** tile provides a summary of the ongoing and completed tuning actions for your database, giving you a quick view into the history of tuning activity. Clicking this tile takes you to the full tuning history view for your database.
- The **Auto-tuning** tile shows the auto-tuning configuration for your database (tuning options that are automatically applied to your database). Clicking this tile opens the automation configuration dialog.
- The **Database queries** tile shows the summary of the query performance for your database (overall DTU usage and top resource consuming queries). Clicking this tile takes you to **Query Performance Insight.**

AUTOMATIC TUNING

- Azure SQL Databases can automatically tune database performance by applying performance recommendations.
- Azure SQL Database is an automatically managed data service that constantly monitors your queries and identifies the action that you can perform to improve performance of your workload.
- You can review recommendations and manually apply them, or let Azure SQL Database automatically apply corrective actions - this is known as **automatic tuning mode**.
- Automatic tuning can be enabled at the server or the database level. Auto Tuning is enabled at the server level from the portal and it is enabled at database level by running the following SQL:

```
ALTER DATABASE current SET AUTOMATIC_TUNING = AUTO | INHERIT | CUSTOM
```